

Students Violation Monitoring System

A System Proposal

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In the Partial Fulfillment

Of the Requirements for

Systems Design and Analysis

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CHAPTER 1

INTRODUCTION

Rapid technological developments have helped people make their lives easier. The field of education has also experienced significant advancement due to the remarkable growth in technology ([Madhumitha et al., 2021](#)). In the age of digitization, academic institutions are using advanced technologies to improve administrative procedures. Since traditional manual methods are often inaccurate and time-consuming ([Rayan et al., 2025](#)). One such innovation is the Student Violation Monitoring System. It is designed to digitalize the manual processes in monitoring students' violations in the Office of the Student Affairs (OSA) of Carlos Hilado Memorial State University (CHMSU). This system is developed using the Django framework, designed to automate the processes of recording, tracking, and monitoring the students' violations within the academic institution. The Office of the Student Affairs (OSA) of Carlos Hilado Memorial State University (CHMSU) uses traditional methods for monitoring violations, which often rely on manual logbooks and paper-based documentation that are prone to human error, inconsistent record-keeping, and significant processing delays. The proposed system aims to provide a centralized database that efficiently stores, updates, and retrieves records related to student violations. By integrating automation and real-time data processing, the system minimizes the possibility of inaccurate or lost records and ensures consistent data access among authorized users.



1.1 Background of the Study

Educational institutions play a vital role in shaping students' academic growth and character development. Alongside academic instruction, universities are responsible for maintaining discipline and ensuring that students adhere to established rules and policies. At Carlos Hilado Memorial State University (CHMSU) – Main Campus, the Office of Student Affairs (OSA) is tasked with managing student conduct and addressing violations to promote a safe, orderly, and conducive learning environment. Currently, the monitoring and documentation of student violations at CHMSU largely rely on manual and paper-based processes, such as logbooks, written incident reports, and physical record filing. While these traditional methods have been functional, they present several challenges, including delayed processing of cases, difficulty in tracking violation history, inconsistent record-keeping, and increased risk of data loss or human error. As student populations grow and administrative responsibilities increase, these limitations hinder the efficiency of disciplinary management and timely intervention.

The advancement of information technology has enabled educational institutions to improve administrative operations through digital systems. The adoption of web-based solutions allows real-time access to records, streamlined communication among stakeholders, and centralized data storage. In line with these developments, the need for an automated system to manage student violations at CHMSU has become increasingly apparent. The Student Violation Monitoring System is proposed to address these challenges by digitalizing the processes involved in reporting, monitoring, and resolving student violations. The system provides a centralized platform where authorized users—such as guards, staff, students, student formators, and the OSA coordinator—can efficiently perform their respective roles. Through automated citation issuance, notification mechanisms, letter of apology processing, and analytical reporting, the system aims to enhance transparency, accuracy, and coordination within the disciplinary process. By implementing this system at Carlos Hilado Memorial State University – Main Campus, the institution can improve the effectiveness of its student discipline management, support informed decision-



making, and strengthen institutional governance. Ultimately, the proposed system contributes to a more organized, accountable, and technology-driven approach to maintaining student discipline within the university.

1.2 Objectives of the Study

The main objective of this study is to develop a web-based Student Violation Monitoring System that digitizes and improves the recording, monitoring, and management of student violations at Carlos Hilado Memorial State University.

Specifically, this study intends to:

- To develop and deploy a functional web-based Student Violation Monitoring System that records and tracks student violations within the Office of Student Affairs by the end of the development period
- To ensure that the system allows authorized users to access and update violation records accurately and consistently throughout the implementation phase
- To improve the efficiency of violation processing by reducing manual record-keeping and processing delays during the system's initial use
- To enable timely dissemination of violation-related information through automated notifications within the system upon implementation
- To generate organized violation reports and summaries that can be utilized by the OSA for monitoring and evaluation at the end of each academic term



1.3 Scope and Limitation

The Student Violation Monitoring System (SVMS) is designed to support the monitoring, documentation, and resolution of student violations within the Office of Student Affairs (OSA) of Carlos Hilado Memorial State University (CHMSU). The system covers the entire violation management process, beginning with the issuance of citation tickets by guards, confirmation and sanction processing by staff, submission and evaluation of letters of apology by students and student formators, and monitoring and reporting by the OSA coordinator. It provides a role-based access interface for five user types: guard, staff, student, student formator, and OSA coordinator, ensuring that each user can only perform authorized actions. The system includes automated notifications, centralized record storage, violation tracking, sanction scheduling, and statistical reporting features such as violation trends, demographic analysis, and case status monitoring. The system operates as a web-based platform developed using the Django framework and is accessible to authorized users within the institution.

Despite its comprehensive features, the Student Violation Monitoring System is limited to managing student violation records within CHMSU and does not extend to external institutions or non-academic disciplinary cases. The system relies on the accuracy of data entered by authorized users; therefore, incorrect or incomplete inputs may affect report reliability. It does not include biometric identification, CCTV integration, or automated violation detection, as all violations must be manually recorded by guards or staff. The system requires a stable internet connection and compatible devices to function properly. Additionally, the system



does not replace existing institutional disciplinary policies but serves only as a digital tool to support and improve existing procedures. Data analysis and generated insights are based solely on stored records and may not reflect factors outside the system's scope.

1.4 Significance of the Study

The Student Violation Monitoring System is significant as it addresses the challenges of managing student discipline through traditional manual processes at Carlos Hilado Memorial State University (CHMSU) – Main Campus. By digitalizing the recording, monitoring, and reporting of student violations, the system promotes efficiency, accuracy, and transparency in disciplinary management.

Office of Student Affairs (OSA), the system serves as a centralized and reliable platform for monitoring student violations and disciplinary cases. It enables the OSA to access organized records, track case statuses, and generate analytical reports that support informed decision-making, policy evaluation, and institutional planning.

Guards, the system provides an efficient tool for recording violations and issuing citation tickets electronically. It reduces reliance on handwritten reports, ensures accurate documentation of incidents, and allows guards to receive timely updates regarding the confirmation of violations and the release of confiscated student IDs.



Staff, the system supports the systematic confirmation of violations and the management of sanctions, penalties, and community service schedules. It enhances coordination with guards, students, and student formators by providing real-time notifications and updated records, thereby reducing delays and administrative workload.

Student Formators, the system offers a structured platform for reviewing and evaluating students' letters of apology. It simplifies the approval, rejection, or revision process and ensures that decisions are properly documented and communicated to both students and staff.

Students, the system increases transparency by allowing them to view their violation records, understand required actions, and submit letters of apology through a secure digital platform. This promotes accountability, awareness of university policies, and timely compliance with disciplinary requirements.

Institution, the system contributes to improved governance by providing historical data and statistical insights on student violations. These insights can be used to identify trends, improve preventive measures, and enhance campus discipline strategies.

Future Researchers and System Developers, this study may serve as a reference for the development of similar information systems related to student discipline management and administrative automation in higher education institutions.



1.5 Definition of Terms

To further clarify the context for the readers of this study, the following terms are defined conceptually and operationally:

Student Violation Monitoring System (SVMS). A web-based information system developed to digitalize the recording, monitoring, and management of student violations at Carlos Hilado Memorial State University.

Office of Student Affairs (OSA). The university office responsible for overseeing student welfare, discipline, and the implementation of policies related to student conduct.

Violation. Any act or behavior committed by a student that is contrary to the rules and regulations of the university.

Citation Ticket. An electronic document issued by guards through the system to formally record a student violation and initiate the disciplinary process.

Guard. An authorized personnel responsible for identifying violations, confiscating student identification cards, and issuing citation tickets within the system.

Staff. Authorized personnel who review and confirm citation tickets, manage violation records, and assign appropriate sanctions or penalties.



Student. An officially enrolled individual at Carlos Hilado Memorial State University who may view violation records and submit a letter of apology through the system.

Student Formator. A designated authority responsible for reviewing, approving, declining, or requesting revisions to students' letters of apology.

OSA Coordinator. An authorized user who oversees the overall violation monitoring process and generates reports and analytical insights through the system.

Letter of Apology. A formal written statement submitted by a student through the system expressing accountability for a violation and requesting consideration from the student formator.

Sanctions or Penalties. Disciplinary actions assigned to a student as a consequence of a violation, such as community service or other corrective measures.

Community Service. A corrective activity assigned to students as part of their sanction, requiring them to render service within the university or designated areas.

Notification. An automated system-generated alert that informs users about updates, actions, or decisions related to violation records.

Dashboard. A visual interface within the system that displays summarized data, statistics, and reports related to student violations.

Major Violation. A serious offense that requires stricter disciplinary action in accordance with university policies.



Minor Violation. A less serious offense that requires minimal disciplinary action as defined by university rules.

Centralized Database. A single, secure data repository where all student violation records and related information are stored and managed.

Web-Based System. A system that is accessible through a web browser and does not require local software installation.



CHAPTER II

REVIEW OF RELATED LITERATURE/STUDIES

This chapter presents a review of related literature, studies, and existing systems that serve as the basis for the development of the Student Violation Monitoring System (SVMS). It provides both theoretical and practical foundations for designing a web-based system that improves the monitoring, recording, and management of student violations in educational institutions. The chapter is organized into foreign and local related studies, a summary of key findings, and a synthesis of related systems, which collectively justify the need for the proposed system.

2.1 Foreign-Related Literature

Student Tracker: Enhancing Safety and Academic Accountability Through Smart Monitoring

Rayan et al., (2025) investigated the development and implementation of a Student Tracking App designed to enhance safety and academic accountability through smart monitoring in educational institutions. Their study detailed how the application, built using Android Java and Firebase, automates student management functions such as attendance tracking, profile handling, and real-time monitoring of student presence through GPS-based location verification. The application restricts



attendance marking to a defined campus zone and operating hours, and sends SMS alerts to administrators for attendance violations, thereby improving accuracy and administrative responsiveness. Additionally, the app supports final-year students by enabling the upload of resumes and academic information, bridging traditional manual processes with modern digital solutions.

ViMoIS: The Student Violation Point Monitoring Information System

Rozy and Purnama (2022) developed ViMoIS, a web-based Student Violation Point Monitoring Information System designed to modernize how student behavioral infractions are recorded and monitored in schools. Their research highlighted that conventional manual recording of student violations is prone to data loss and delays and often leaves parents unaware of their children's behavior in school. To address this, they used the waterfall model of the System Development Life Cycle (SDLC) to analyze, design, and test the system, which enables digital recording of violation points and allows parents to monitor their children's conduct online. ViMoIS includes key features such as automated violation recording, WhatsApp notifications to parents, and a live chat function between counseling teachers and parents, ensuring more efficient communication and transparency in the disciplinary process. Black-box testing showed that all components of the system operated successfully, demonstrating that digital systems can replace traditional manual methods effectively.

Web-Based Student Violation Monitoring Information System Design at SMK Gandasari



Triansyah et al. (2022) designed a Web-Based Student Violation Monitoring Information System to improve how student disciplinary violations are recorded and communicated at SMK Gandasari Vocational School. Their study noted that traditional paper-based recording of student violations made it difficult for parents to receive timely information about their children's behavior, as reports were only shared when report cards were distributed. To address this, they developed a web-based system using the Waterfall development model, which includes stages like requirements analysis, design, implementation, testing, and maintenance. The system uses Unified Modeling Language (UML) diagrams for planning and supports features such as violation data entry, user management, and generation of violation reports that teachers and parents can access more easily in real time. Results from black-box testing confirmed that the system's components function as intended, demonstrating that web-based solutions can make monitoring student discipline more efficient and transparent compared with manual methods.

2.2 Local Related Literature

Yellow Form 2.0: A Digital Monitoring System for Student Decorum Violations

Pugeda (2025) investigated the development of Yellow Form 2.0, a web-based digital monitoring system created to replace traditional manual reporting of student decorum violations at St. Paul University Philippines. Historically, the "yellow form" was used as a paper-based means for documenting student behavior infractions, which often resulted in delayed tracking, scattered records, and



difficulties in identifying recurring issues across departments. To improve this process, Yellow Form 2.0 allows faculty and staff to digitally file violation reports, monitor case status, and view behavior trends through a centralized platform, thereby enhancing administrative efficiency and real-time visibility into student conduct. The system was developed using Agile Scrum methodology and evaluated against ISO/IEC 25010 quality standards for functionality, usability, reliability, and security, demonstrating its suitability for modern behavior tracking and administrative workflows.

Student ID Validation Monitoring System

Muelan, Decomotan, and Lebuna (2024) developed the Student ID Validation Monitoring System (SIVMS), a web-based monitoring solution designed to ensure that only students with validated identification cards are present on campus at Iloilo Science and Technology University, Miagao Campus. The system uses QR code scanning technology to quickly and accurately verify student ID status, addressing challenges faced by the Office of Student Affairs and Services in maintaining campus safety and security. Built with HTML, CSS, PHP, and MySQL and developed using an Iterative Software Development Methodology, the system also includes features for managing student profiles, generating unique QR codes, and generating reports. Evaluation results based on the ISO/IEC 25010 quality model showed an overall rating of “excellent,” meeting standards for functionality, performance, usability, reliability, security, maintainability, and portability, thereby demonstrating its effectiveness in improving institutional monitoring processes.



The Design and Development of Student Information and Violation Management System (SIVMS)

Olipas (2020) investigated the design and development of a Student Information and Violation Management System (SIVMS) for a higher educational institution in Central Luzon, Philippines. The study utilized a descriptive-developmental research design and applied the Incremental Model of the Software Development Life Cycle (SDLC) to guide the process of analysis, design, development, and testing of the prototype system. The research detailed how stakeholder interviews and document analysis were used to capture requirements for organizing student data and violation records, which were then translated into system models such as use-case diagrams, data flow diagrams, and entity-relationship diagrams. The prototype aided in digitalizing what was previously a manual, paper-based process of managing student information and violations, intending to improve efficiency, accuracy, and the organization of records. The study concluded with recommendations for further development and deployment of the system in educational settings.

2.3 Synthesis

TABLE I

Syntheses of Foreign Related Systems and Literature

Foreign Related Systems and Literature	Features and Functionalities	Developed System (Student Violation Monitoring System)
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Student Tracker: Enhancing safety and Academic Accountability Through Smart Monitoring (Rayan et al., 2025)	Attendance tracking, profile management, real-time GPS monitoring, automated alerts	SVMS will track student violations digitally, with real-time notifications to staff and OSA Coordinator
ViMoIS: The Student Violation Point Monitoring Information System (Rozy, A. F., & Purnama, R. 2022)	Automated violation recording, online monitoring by parents, WhatsApp alerts, live chat with teachers	SVMS will allow digital recording of violations, accessible by guard, staff, student, student formator, and OSA Coordinator with alert notifications
Web-Based Student Violation Monitoring Information System Design at SMK Gandasari (Triansyah et al., 2022)	Violation data entry, user management, violation report generation, real-time access	SVMS will include violation entry, report generation, and real-time access to improve transparency and efficiency

TABLE II

Syntheses of Local Related Systems and Literature

Local Related Systems and Literature	Features and Functionalities	Developed System (Student Violation Monitoring System)
Yellow Form 2.0: A digital monitoring system for student decorum violations (Pugeda, R. D. 2025)	Web-based violation reporting, tracking case status, centralized behavior trends	SVMS will enable web-based violation logging, track cases, and analyze behavior trends
Student ID Validation Monitoring System (Muelan et al., 2024)	QR code ID validation, student profile management, report generation	SVMS will integrate student identification and violation tracking, ensuring accurate records
Student Information and Violation Management	Digital record of student data and violations, use-	SVMS will automate the organization and



System (Olipas, 2020)	case and data flow diagrams, system testing	management of student violation data for efficiency and accuracy
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CHAPTER III

Methodology

This chapter presents the design and methodology used to develop the Student Violation Monitoring System (SVMS). It identifies the appropriate research approach to select a compatible Software Development Life Cycle (SDLC) model and outlines the system's requirements specification to guide its design and development.

3.1 Systems Development Method

The systems development method refers to the framework used to structure, plan, and control the process of developing the Student Violation Monitoring System for CHMSU. The proponents used the Agile methodology, specifically an incremental development model, which allows flexibility and continuous feedback throughout the system development process. This approach enables the researchers to build the system in small, manageable parts, ensuring that features for tracking student offenses are refined and improved based on the actual needs of the university's disciplinary office.



Fig. 1. Agile Methodology Model

Planning

In the Planning phase, the researchers collaborated with the Office of Student Affairs (OSA) of CHMSU to define the project's scope, goals, and necessary features. The process began with the submission of a formal request letter to the OSA for administrative approval. Once cleared, the team conducted interviews and gathered data from the office to ensure all relevant information regarding student offenses, penalties, and sanctions was accurately collected. Additionally, the researchers defined the project's objectives and requirements, ensuring that the Student Violation Monitoring System would effectively meet the operational needs of the university's disciplinary staff.

Design

During the Design phase, the gathered requirements were translated into the system architecture and wireframes. The researchers developed the system's architecture to ensure it supported the specific administrative workflows of the Office of Student Affairs (OSA). They designed the database schema to handle the data



needs of the university, ensuring the system could efficiently store and manage records related to student profiles, categorized offenses, and the history of applied sanctions. The user interface was designed to be intuitive and efficient, with input from OSA to ensure that the system is easy to navigate for the Student, Guard, Formator, Coordinator and Staff.

Develop

The Development phase was executed through a series of short, iterative sprints, typically lasting one to two weeks per cycle. During these sprints, the researchers focused on building the highest-priority modules from the product backlog, utilizing continuous feedback from the Office of Student Affairs (OSA) to refine the system's core features. The development team set up the development environment, ensuring it was tailored to the administrative requirements of the university. The team then implemented the centralized database and developed the front-end interface, ensuring it was user-friendly for staff managing student records. The back-end functionalities were developed to support the automated tracking of student offenses, and the integration of the notification and reporting module was completed to enable real-time updates on sanctions and the generation of disciplinary reports for university officials.

Test



The Test phase ran concurrently with development to ensure every new feature worked correctly. The team conducted unit testing of individual components, followed by integration testing to ensure the system as a whole functioned smoothly. User Acceptance Testing was carried out with the Students of Information System in CHMSU to gather feedback on usability, functionality, and performance. The researchers utilized the ISO/IEC 25010:2011 standard to assess the system's quality in terms of its functional suitability, performance efficiency, and usability. Additionally, ISO 9241-11:2018 was applied to measure user comfort and satisfaction during system navigation. Based on the gathered feedback, bugs and errors were fixed, and optimizations were made to ensure the system effectively manages the university's disciplinary records.

3.2 Operational Framework

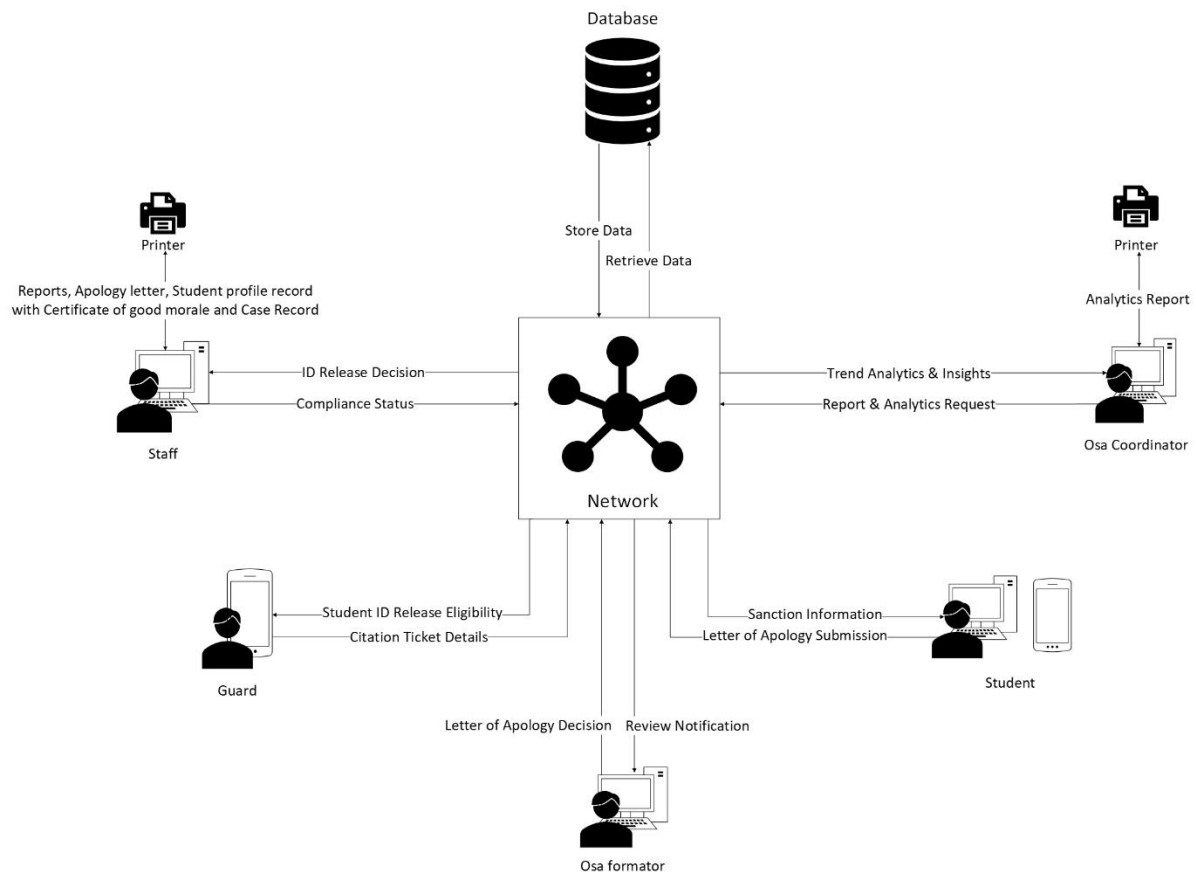


Fig. 2. Operational Framework of Student Violation Monitoring System

Figure 2 shows the operational framework of the Student Violation Monitoring System (SVMS) for CHMSU. It illustrates how the OSA Coordinator, OSA Formator, Staff, Guards, and Students interact with the system to input and access disciplinary information. The OSA Coordinator views trend analytics and insights to monitor campus discipline. The OSA Formator manages the review process by providing review notifications and making final decisions on letters of apology submitted by students. Staff handle the input of compliance status and receive decisions regarding ID releases, ensuring administrative records are up to date. Guards input citation ticket details directly into the system and receive real-time updates on a student's ID



release eligibility. Students access the system to view their specific sanction information and submit their required letters of apology for review. All information is stored in a central database, making it easy to access and manage student records. The printer is used to produce hard copies of analytics reports, apology letters, student profiles, and case records. This system helps improve disciplinary management, making university operations faster, more organized, and responsive to the needs of the Office of Student Affairs, the faculty, and the student body.

3.3 Context Diagram

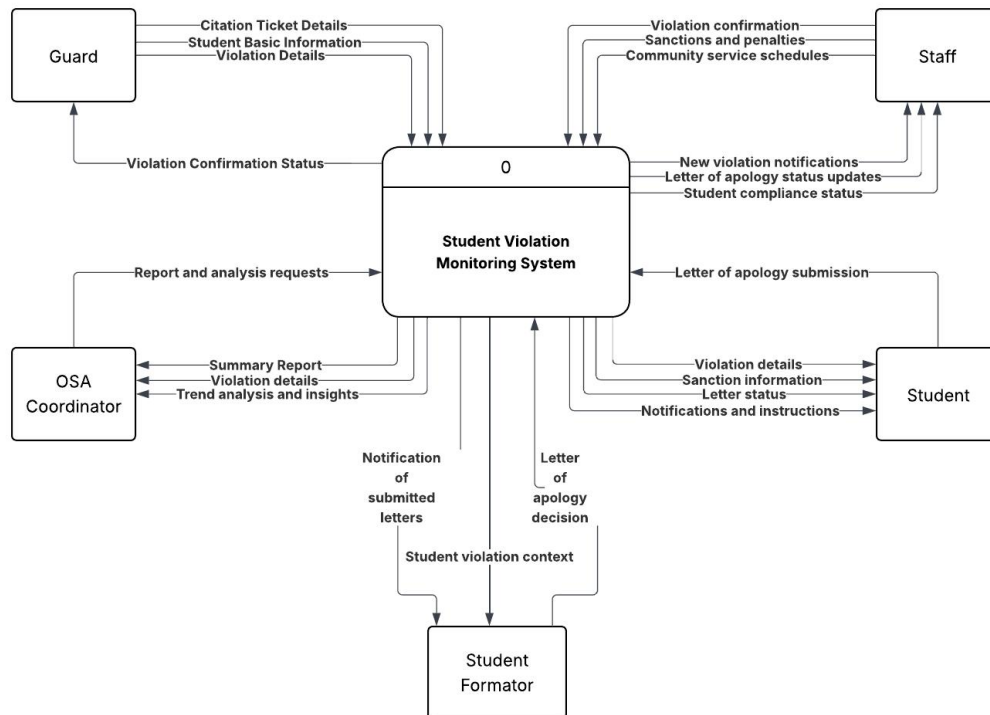


Fig. 3. Context Diagram (Level 0) Of Student Violation Monitoring System.

Figure 3 shows the Context Diagram of the Student Violation Monitoring System (SVMS) illustrates the system as a single unified process interacting with five external entities: Guard, Staff, Student, Student Formator, and OSA Coordinator. The Guard submits citation tickets and violation details to the system and receives confirmation and ID release notifications. Staff interact with the system by validating violations, assigning sanctions, and making release decisions while receiving updates on student compliance. Students access the system to view violations, submit letters of apology, and receive notifications regarding their case status. The Student Formator evaluates submitted apology letters and sends approval or revision decisions through the system. Lastly, the OSA Coordinator accesses the

system for monitoring purposes, generating statistical reports, analyses, and decision-support insights. All interactions

3.4 Data Flow Diagram

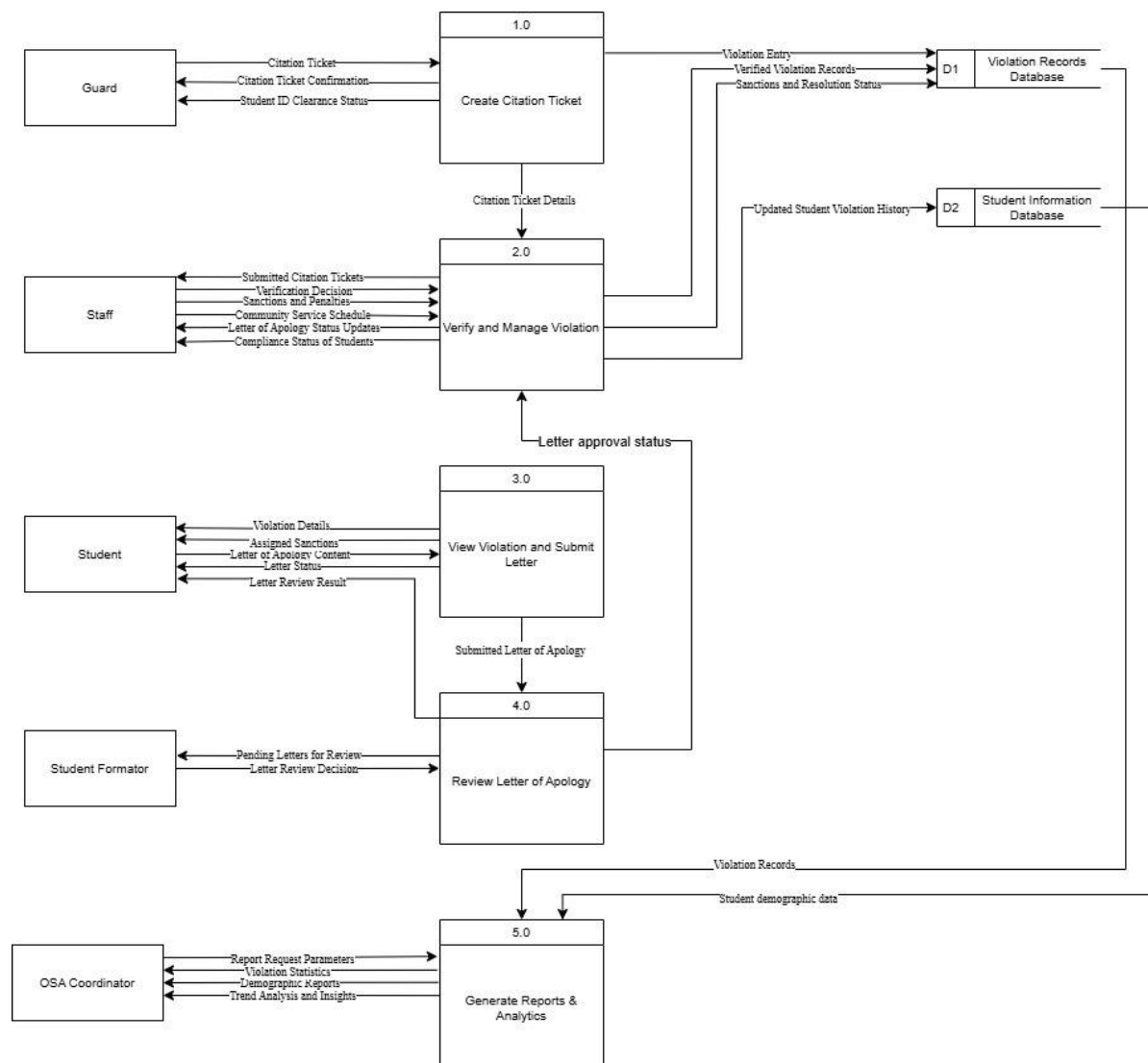


Fig. 4. Data Flow Diagram Level 1 of the Student Violation Monitoring System

Figure 4 shows the Level 1 Data Flow Diagram (DFD) of the system, illustrating the detailed data processes, data stores, and interactions among the system's users. It expands the context diagram into specific modules such as



Manage User, Login, Profiling, Geo Mapping, Service, SMS, Monitoring, and Reports, and presents how each office (SUPER ADMIN, OSCA, PWD Affairs Office, and LYDO) exchanges information with these processes. The diagram also highlights all incoming and outgoing data flows, including account information, profiling records, service data, geolocation information, and generated reports, to provide a clearer view of the system's internal operations.

3.5 Entity Relationship Diagram

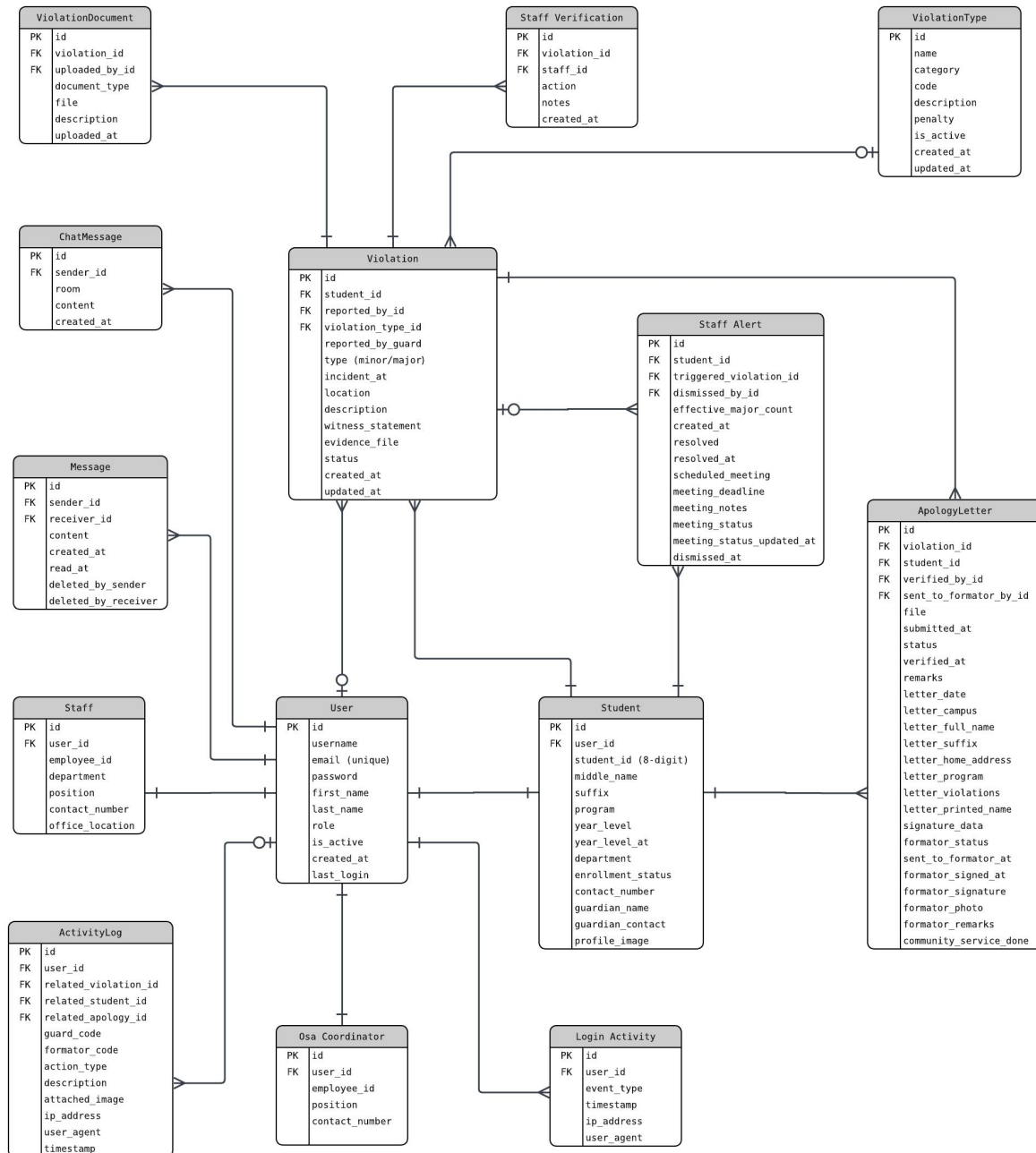


Fig. 5. Entity Relationship Diagram of Student Violation Monitoring System



Figure 5 presents the Entity Relationship Diagram (ERD) of the Office of Student Affairs (OSA) Violation Management and Monitoring System. The diagram illustrates how student violation records are stored, processed, and interconnected with users, staff, and administrative actions within the system.

At the core of the ERD is the Violation entity, which captures detailed information about reported incidents, including the student involved, violation type, incident details, status, and supporting evidence. This entity is linked to ViolationType, which defines the classification, category, and corresponding penalties for each violation, ensuring standardized handling of offenses.

The system also integrates user management through the User entity, which is associated with specialized roles such as Student, Staff, and OSA Coordinator. These entities store role-specific information while maintaining a centralized user account structure for authentication and access control. Communication within the system is supported by the Message and ChatMessage entities, enabling direct interaction between users.

Administrative processes are further represented by entities such as Staff Verification, Staff Alert, and ApologyLetter, which manage verification actions, disciplinary notifications, meeting schedules, and formal student responses. Supporting entities like ViolationDocument allow the attachment of evidence and related files, while ActivityLog and LoginActivity record system actions and access events for auditing and monitoring purposes.

Overall, this ERD provides a structured view of how data flows within the system, ensuring efficient violation tracking, accountability, secure access, and

comprehensive documentation. The design supports effective monitoring, decision-making, and enforcement of student discipline policies within the institution.

3. Use Case Diagram

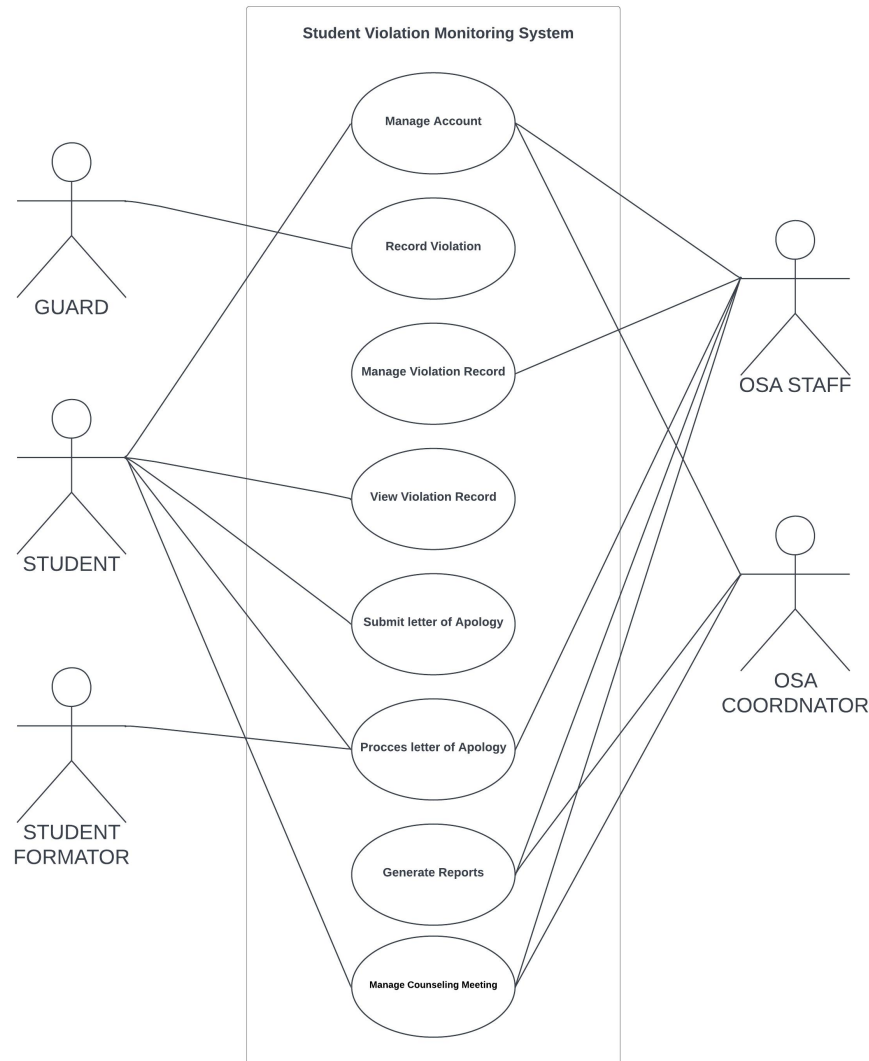


Fig. 6. Use Case Diagram of Student Violation Monitoring System

Figure shows the UML use case diagram for the Student Violation Monitoring System (SVMS), which illustrates the functional interactions among five (5) primary actors: the Guard, Student, OSA Staff, Student Formator, and OSA Coordinator. The



diagram outlines the core operational tasks of the guard, including recording student violations, while highlighting student-related functions such as viewing violation records and submitting letters of apology. It also identifies shared system functions such as account management and violation record viewing, which are accessible to authorized users. Furthermore, the diagram defines the administrative and decision-support roles of the OSA Staff and OSA Coordinator, who are responsible for managing violation records, processing apology letters, and generating statistical reports for monitoring and policy evaluation.



3.7 Data Dictionary

Table 3

Users

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique identifier for each user
username	Varchar	255	—	Unique username for login
email	Varchar	255	—	Unique email address of the user
password	Varchar	255	—	Encrypted password for authentication
first_name	Varchar	255	—	Legal first name of the user
last_name	Varchar	255	—	Legal last name of the user
role	Varchar	50	—	User role (e.g., Student, Staff, OSA, Admin)
Is_active	Boolean	1	—	Status to check if the account is enabled
Created_at	Datetime	—	—	Date and time the user was created
last_login	Datetime	—	—	Timestamp of the most recent login



Table 4

Student

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique identifier for the student record
user_id	Integer	11	Foreign Key	Reference to the User table
student_id	Varchar	8	—	8-digit unique institutional student ID
middle_name	Varchar	255	—	Student's middle name
suffix	Varchar	10	—	Name suffix (e.g., Jr., III)
program	Varchar	255	—	Academic program/course
year_level	Integer	2	—	Current academic year level
year_level_at	Datetime	—	—	Timestamp of when year level was updated
department	Varchar	255	—	Academic department
enrollment_status	Varchar	50	—	Status (e.g., Enrolled, Dropped, Graduated)
contact_number	Varchar	50	—	Student's phone number
guardian_name	Varchar	255	—	Name of parent or guardian
guardian_contact	Varchar	50	—	Phone number of parent or guardian
profile_image	Varchar	255	—	Path/URL to the student's profile photo



Table 5
Violation

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique identifier for each violation incident
student_id	Integer	11	Foreign Key	Reference to the Student table
reported_by_id	Integer	11	Foreign Key	Reference to the User who reported it
violation_type_id	Integer	11	Foreign Key	Reference to the ViolationType table
reported_by_guard	Boolean	1	—	Indicates if the reporter was a guard
type	Varchar	10	—	Category: minor or major
incident_at	Datetime	—	—	Date and time the incident occurred
location	Varchar	255	—	Specific location of the incident
description	Text	—	—	Detailed account of the violation
status	Varchar	50	—	Current status (e.g., Pending, Resolved)
created_at	Datetime	—	—	Date the record was created



Table 6
Violation Type

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique ID for the violation category
name	Varchar	255	—	Name of the violation type
category	Varchar	100	—	Broad category of the offense
code	Varchar	50	—	Standard code for the offense
penalty	Text	—	—	Prescribed penalty for this violation
Is_active	Boolean	1	—	Indicates if this type is still in use



Table 7
Apology Letter

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique identifier for the letter
violation_id	Integer	11	Foreign Key	Reference to the related Violation
student_id	Integer	11	Foreign Key	Reference to the Student author
status	Varchar	50	—	Status (e.g., Submitted, Verified)
letter_full_name	Varchar	255	—	Printed full name on the letter
signature_data	Text	—	—	Digital signature or path to image
community_service_done	Boolean	1	—	Completion status of required service



Table 8
Activity Log

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique ID for the log entry
user_id	Integer	11	Foreign Key	Reference to the User performing action
action_type	Varchar	100	—	Type of action (e.g., CREATE, UPDATE)
description	Text	—	—	Detailed log message
ip_address	Varchar	45	—	Network IP address of the user
timestamp	Datetime	—	—	Date and time of the activity



Table 9

Staff

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique number for each staff record
user_id	Integer	11	Foreign Key	Reference to the User table
employee_id	Varchar	100	—	Unique institutional employee ID
position	Varchar	255	—	Official job title of the staff
contact_number	Varchar	50	—	Staff's primary phone number
office_location	Varchar	255	—	Building/Room assigned to the staff

Table 10

OSA Coordinator

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique ID for the OSA Coordinator
user_id	Integer	11	Foreign Key	Reference to the User table
employee_id	Varchar	100	—	Institutional employee identifier
position	Varchar	255	—	Specific role within the OSA office



Table 11
Staff Verification

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique ID for the verification record
violation_id	Integer	11	Foreign Key	Reference to the Violation being verified
staff_id	Integer	11	Foreign Key	Reference to the Staff performing verification
Action	Varchar	100	—	Decision made (e.g., Approved, Denied)
notes	Text	—	—	Feedback or remarks from the staff
created_at	Datetime	—	—	Date of verification

Table 12
Staff Alert

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique number for each system alert
student_id	Integer	11	Foreign Key	Reference to the flagged Student
triggered_violation_id	Integer	11	Foreign Key	Violation that caused the alert
effective_major_count	Integer	11	—	Running tally of major violations
meeting_status	Varchar	50	—	Status of disciplinary meeting
scheduled_meeting	Datetime	—	—	Date set for the student conference



Table 13

Violation Document

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique ID for the attached file
violation_id	Integer	11	Foreign Key	Linked violation incident
uploaded_by_id	Integer	11	Foreign Key	Reference to the User who uploaded it
file	Varchar	255	—	Path or URL to the stored document
document_type	Varchar	50	—	Category (e.g., PDF, Image, Scan)

Table 14

Message

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique identifier for the message
sender_id	Integer	11	Foreign Key	Unique identifier for the message
receiver_id	Integer	11	Foreign Key	User ID of the recipient
content	Text	—	—	Body of the message
read_at	Datetime	—	—	Timestamp when message was viewed



Table 15
Chat Message

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique ID for chat record
sender_id	Integer	11	Foreign Key	User ID of the sender
room	Varchar	100	—	Identifier for the chat session/room
content	Text	—	—	Textual content of the chat
created_at	Datetime	—	—	Timestamp of the chat message

Table 16
Login Activity

Field Name	Data Type	Length	Key	Description
id	Integer	11	Primary Key	Unique ID for login session
user_id	Integer	11	Foreign Key	Reference to the User table
event_type	Varchar	50	—	Type of event (e.g., login, logout)
timestamp	Datetime	—	—	Date and time of the event
ip_address	Varchar	45	—	Network IP used for the session
user_agent	Varchar	255	—	Browser/Device information



3.7 Hardware and Software Requirements

The proper operation of the Student Violation Monitoring System (SVMS) requires minimum and recommended hardware and software specifications to ensure reliable system performance, data security, and accessibility within the educational institution. Meeting these requirements enables guards, staff, formators, and the Office of Student Affairs (OSA) to efficiently record violations, review cases, generate reports, and monitor student discipline during daily system use.

a) Software Requirements

This section presents the software requirements necessary for the proper operation and daily use of the Student Violation Monitoring System. These requirements ensure that authorized users can reliably access the system, manage violation records, review student cases, and generate statistical and analytical reports.



Table 17

Software requirement for Student Violation Monitoring System

Category	Operational Requirement
Operating System	Windows 10 or higher
Web Server Environment	Cloud-based web hosting platforms (PythonAnywhere, Render); Django Development Server for local testing
Database Management System	PostgreSQL for secure and reliable data storage and management
Web Browser	Google Chrome, Mozilla Firefox, Microsoft Edge
Network Requirement	Local Area Network (LAN) or Internet connection
Programming Languages	HTML, CSS, JavaScript, SCSS, PYTHON
Security Support	Django authentication, role-based access control, and browser cookie support



b) Hardware Requirements

This section outlines the minimum and recommended hardware specifications required to support efficient, secure, and uninterrupted system operation within the school or university environment.

Table 18

Hardware requirements for Student Violation Monitoring System

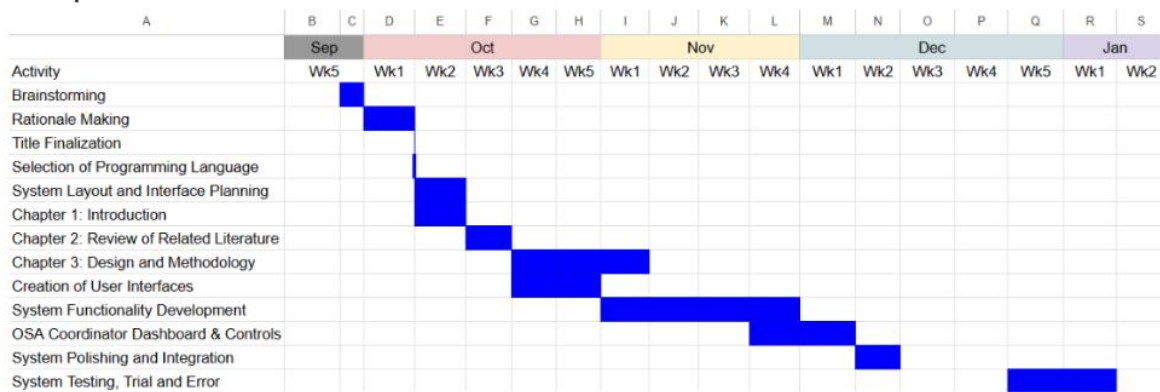
Component	Minimum Requirement	Recommended Requirement
Processor	Processor from Intel i3 or higher	Ryzen AMD 3 and higher
Memory (RAM)	8 GB	16 GB or higher
Storage	100 GB available disk space	SSD with sufficient capacity
Network Device	Ethernet or Wi-Fi adapter	Stable broadband-capable network device
Peripheral Devices	Keyboard, mouse, monitor	Printer for reports and official documents
Devices	Desktop or laptop and Mobile Phone	Desktop or laptop with SSD Android or iOS smartphone with updated OS



3.9 Timetable (Gantt Chart)

Table 19

Student Violation Monitoring System Gantt Chart table



This table presents a visual project timeline in the form of a simplified Gantt chart, illustrating the planned schedule of activities over a five-month project duration from September to January. The timeline is organized sequentially according to the major phases of system development, beginning with early planning and progressing through design, development, integration, and testing. Initial activities such as brainstorming, rationale making, title finalization, and programming language selection are concentrated in the early weeks of the project, establishing a strong foundation. Documentation-related tasks, including Chapter 1 (Introduction) and Chapter 2 (Review of Related Literature), are scheduled during the early to mid stages, followed by Chapter 3 (Design and Methodology), which overlaps with interface creation and system development. Core development activities, such as



user interface creation, system functionality development, and dashboard and control implementation, span multiple weeks and overlap to reflect iterative development. The later stages focus on system polishing, integration, and comprehensive testing through trial and error. Overall, the Gantt chart provides a clear visual overview of task sequencing, durations, and overlaps, highlighting effective workload distribution and steady progress toward project completion.

Table 20

Student Violation Monitoring System Gantt PERT table

Activity Code	Activity	Predecessors	O	M	P	Estimated Time (Days)
A	Brainstorming	—	2	3	4	3
B	Rationale Making of Proposed Title	A	2	3	4	3
C	Finalization of System Title	B	1	1	1	1
D	Selection of Programming Language	C	1	2	3	2
E	System Layout and Interface Planning	D	5	8	11	8



F	Chapter 1: Introduction	E	5	7	9	7
G	Chapter 2: Review of Related Literature	F	5	6	8	6
H	Chapter 3: Design and Methodology	G	14	20	26	20
I	Creation of Coordinator, Staff, Guard and Student Interfaces	H	10	15	30	15
J	System Functionality Development	I	20	30	40	30
K	Creation of Osa Coordinator Dashboard and Controls	J	5	8	11	8
L	System Polishing and Feature Integration	J,K,H	4	6	8	6
M	System Testing, Trial and Error	L	7	10	13	10



Table 21

Slack Time Analysis

Activity Code	Task Name	Duration	Early Start	Early Finish	Late Start	Late Finish	Total Slack	Critical
A	Brainstorming	3	0	3	0	3	0	TRUE
B	Rationale Making of Proposed Title	3	3	6	3	6	0	TRUE
C	Finalization of System Title	1	6	7	6	7	0	TRUE
D	Selection of Programming Language	2	7	9	7	9	0	TRUE
E	System Layout and Interface Planning	8	9	17	9	17	0	TRUE
F	Chapter 1: Introduction	7	17	24	17	24	0	TRUE
G	Chapter 2: Review of Related Literature	6	24	30	24	30	0	TRUE
H	Chapter 3: Design and Methodology	20	30	50	83	103	53	FALSE
I	Creation of Coordinator, Staff, Guard and Student Interfaces	15	50	65	50	65	0	TRUE



J	System Functionality Development	30	65	95	65	95	0	TRUE
K	Creation of Osa Coordinator Dashboard and Controls	8	95	103	95	103	0	TRUE
L	System Polishing and Feature Integration	6	103	109	103	109	0	TRUE
M	System Testing, Trial and Error	10	109	119	109	119	0	TRUE

The PERT/CPM analysis shows that the project is expected to be completed in 119 days. The majority of project activities lie on the critical path, indicating that any delay in these tasks will directly affect the overall project completion time. Critical activities include the initial planning stages, system design, interface creation, core system development, dashboard implementation, system integration, and final testing. Only Chapter 3: Design and Methodology (Activity H) has significant slack, allowing it to be delayed without impacting the project's final deadline. This provides limited scheduling flexibility, particularly for documentation-related tasks, while development and implementation activities require strict adherence to the schedule. Close monitoring, effective coordination, and timely execution of critical activities are essential to ensure that the project is completed within the planned duration.



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